

BORZYA, Russian FederationWMO: **309650**Lat: **50.400N**Lon: **116.517E**Elev: **2217**StdP: **13.56**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.8	-34.5	-42.6	0.5	-38.2	-38.5	0.7	-34.3	20.1	3.0	17.6	1.4	2.3	70	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	88.2	63.7	84.5	62.5	80.8	61.7	67.4	80.2	65.9	77.5	64.4	75.2	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	96.1	71.1	62.2	90.8	69.8	60.6	85.8	68.7	33.2	80.6	32.0	77.5	30.8	75.1	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.1	16.5	DB	-42.9	93.5	5.4	5.2	-46.9	97.2	-50.0	100.3	-53.1	103.2	-57.1	106.9	
			WB	-42.9	70.6	5.4	2.1	-46.7	72.1	-49.9	73.3	-52.9	74.5	-56.7	76.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	-13.8	-5.1	14.0	35.1	49.7	62.7	67.3	62.6	49.7	32.0	8.7	-9.0
	DBStd	29.87	9.25	10.54	12.25	9.00	8.23	6.99	5.35	6.02	7.96	9.23	11.86	9.50
	HDD50	8933	1978	1544	1117	454	105	3	0	2	100	560	1240	1830
	HDD65	13107	2443	1964	1582	898	480	121	32	116	464	1022	1690	2295
	CDD50	1510	0	0	0	6	96	384	538	393	90	3	0	0
	CDD65	209	0	0	0	0	6	52	105	42	4	0	0	0
	CDH74	2963	0	0	0	8	148	883	1253	589	82	0	0	0
	CDH80	963	0	0	0	1	38	310	429	171	14	0	0	0
Wind	WSAvg	6.2	3.8	4.7	6.8	8.9	9.0	6.8	6.2	6.1	6.7	6.6	5.1	3.9
Precipitation	PrecAvg	11.7	0.1	0.1	0.1	0.3	0.7	1.9	3.6	3.0	1.5	0.4	0.2	0.2
	PrecMax	17.0	0.2	0.6	0.6	1.1	2.8	5.0	7.2	7.6	4.6	1.9	0.5	0.4
	PrecMin	6.2	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.9	0.0	0.0	0.0	0.0
	PrecStd	2.8	0.0	0.1	0.1	0.3	0.6	1.2	1.6	1.4	1.1	0.4	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	31.3	52.3	73.0	85.1	94.2	94.1	89.8	81.7	66.5	43.1	21.3
		MCWB	14.9	25.7	39.0	49.2	58.2	63.2	65.1	65.0	58.2	48.4	34.3	18.8
	2%	DB	10.5	23.9	44.6	64.3	78.0	87.2	88.7	84.9	75.7	58.6	35.9	13.8
		MCWB	9.0	20.3	34.0	45.5	54.6	61.4	64.9	63.3	56.3	43.7	29.5	12.2
	5%	DB	5.9	18.7	39.4	58.3	72.7	83.1	84.9	80.5	70.5	53.1	31.3	9.7
		MCWB	4.7	16.2	30.9	42.4	52.1	60.3	64.4	62.7	54.1	41.0	26.8	8.4
	10%	DB	1.8	13.5	34.1	53.0	67.7	78.8	80.8	76.3	65.9	48.1	26.5	6.1
		MCWB	0.8	11.7	27.7	39.5	49.7	59.6	63.5	61.4	52.1	38.4	23.1	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	25.5	39.3	50.4	60.9	66.8	70.2	68.9	61.5	48.7	34.8	18.6
		MCDB	16.7	30.4	51.5	70.9	75.7	84.6	81.6	82.4	74.2	65.7	42.5	21.1
	2%	WB	8.9	20.3	34.5	46.3	56.4	64.5	68.2	66.4	58.9	44.6	30.3	12.3
		MCDB	10.3	23.8	43.7	62.1	74.0	78.8	80.9	76.8	69.4	56.1	35.4	13.8
	5%	WB	4.7	16.1	31.0	43.1	53.3	62.9	66.7	64.7	56.4	41.8	26.8	8.4
		MCDB	5.8	18.6	38.6	57.1	70.7	76.8	79.0	74.8	67.6	51.9	31.2	9.6
	10%	WB	0.9	11.8	28.0	39.9	50.8	61.4	65.5	63.2	53.6	39.0	23.1	5.1
		MCDB	1.8	13.5	34.1	52.2	65.9	74.8	76.7	72.9	64.5	47.6	26.3	6.1
Mean Daily Temperature Range	5% DB	MDBR	24.3	28.1	26.9	25.1	26.0	24.4	21.9	22.5	24.9	22.9	22.5	23.1
		MCDBR	26.6	30.7	30.9	34.8	34.5	32.6	29.7	30.6	32.9	31.4	25.1	24.7
		MCWBR	25.1	27.3	21.6	19.9	16.7	12.7	10.9	12.6	16.1	18.9	19.3	23.1
	5% WB	MCDBR	26.3	30.7	29.7	33.1	30.8	25.2	22.2	21.7	25.9	28.4	24.9	24.6
		MCWBR	24.9	27.4	21.2	19.4	15.6	11.0	9.7	10.5	13.2	17.5	19.5	23.1
Clear-Sky Solar Irradiance	taub	0.253	0.279	0.320	0.410	0.484	0.467	0.421	0.388	0.343	0.311	0.281	0.260	
	taud	2.226	2.172	2.154	2.040	1.883	2.050	2.258	2.324	2.389	2.379	2.274	2.188	
	Ebn at Noon	250	270	278	262	246	252	264	267	269	257	234	224	
	Edh at Noon	22	31	38	49	60	51	41	36	30	25	21	20	
All-Sky Solar Radiation	RadAvg	510	873	1355	1688	1922	1981	1838	1643	1301	896	524	382	
	RadStd	23	40	57	96	131	173	117	113	89	45	26	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	-4.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page